

Form-Induced Birefringence in Elliptical Hollow Photonic Crystal Fiber With Large Mode Area

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Abstract—We propose a novel type of photonic crystal fiber (PCF), including an elliptical hole in its solid core. We prove the feasibility of such a fiber and investigate both experimentally and theoretically the dependence of its group birefringence on the geometric hole parameters. We show, for the first time, that form-induced birefringence can be achieved in single mode PCFs with large mode area and suggest it as a possible route for the development of polarization maintaining PCF-based LMA fiber devices.

Index Terms—Birefringence, fiber characterization, fiber design and fabrication, large core fiber, photonic crystal fiber, single mode fiber.

I. INTRODUCTION

ONGOING research on photonic crystal fibers (PCFs) [1] includes applications in a wide range of areas as optical signal transmission [2], [3], high-power lasers [4], nonlinear fiber optics [5], gas-based nonlinear optics [6], optical signal processing [7], and others. This versatility is due to the particular design flexibility of PCFs which allows them to fit a specific application by varying its geometrical characteristics and structure.

For instance, in the simplest scheme, PCFs are constituted of an array of air holes running along the entire fiber length. If the central hole of the array is replaced by a silica core, these PCFs guide light by means of a *modified* form of total internal reflection and they can show single mode operation with a very high or very low nonlinearity, depending on their geometric features [1].

In a different structure design, hollow core PCFs (HC-PCFs) can also be made by enlarging the central hole of the periodic air hole array. This class of PCFs allows the guidance of light in air, by means of a photonic bandgap effect [1], and open up novel possibilities in both gas-based optics [6] and optical communications [2].

An interesting feature of solid core PCFs is their employment in the realization of high-power lasers [4]. Indeed, PCF technology allows single mode operation over a wide wavelength range when some particular geometric conditions are respected, i.e., the ratio between the hole size d and the hole pitch Λ is less than about 0.43 [8]. This characteristic of PCFs does not depend

on the particular fiber core size but is scalable, in principle, to any core size. Therefore, single mode PCFs with very large mode area (LMA) can be realized for the generation or the delivery of very high optical power [9]. The realization of dual-clad LMA-PCF lasers [4] is particularly advantageous because the use of an air outer cladding avoids any direct contact of radiation to the coating material and leads to a very high numerical aperture (NA) for the inner cladding, providing efficient pump launching, reduced pump threshold, as well as high pump absorption.

Polarization maintenance is a critical issue in clad pumped fiber laser cavities, along with single mode and single polarization operation [10]. Thus, the realization of highly birefringent LMA-PCF amplifiers is of crucial importance.

Several polarization maintaining (PM) solid core PCFs with small core size have been designed and fabricated by several research groups. All of them are primarily based on form birefringence which is obtained by enlarging asymmetrically the fiber core or modifying some hole sizes, shapes or positions [11]–[14]. Indeed an asymmetric core design, combined with the large refractive index contrast achievable in PCFs and the small mode area, allows obtaining a birefringence in the order of 10^{-3} , at least one order of magnitude greater than in conventional highly birefringent fibers [15]. However, it has been shown that, when the core size enlarges, the birefringence of these fibers decreases rapidly [11]. Thus, form-induced birefringence has been considered a valid method for making PM fibers in the small core regime only.

A different method, based on stress-induced birefringence, has been used to make polarization maintaining PCFs with large mode area [16]. In that work the outer cladding of a LMA-PCF was filled with some stress-applying parts (SAPs), similarly to conventional PANDA fibers. These parts are normally made of a material with a different expansion coefficient than that of silica, which can provide a stress field in the fiber after fiber drawing [15]. By using this method, a close to wavelength independent birefringence, in the order of 1.5×10^{-4} , was obtained in a PCF with a calculated mode field diameter (MFD) of $7.1 \mu\text{m}$ [16]. (Note that in the LMA regime the MFD is almost wavelength independent [17].) Recently, better results have been obtained by the same group and single mode PM-PCFs have been commercialized with a core size up to $16 \mu\text{m}$.¹ However, in principle, single mode operation can be obtained in PCFs with a much larger core size. For example, in [18], a ytterbium-doped LMA PCF was fabricated with a core size of about $40 \mu\text{m}$ and an outstanding effective area of about $1000 \mu\text{m}^2$. In [19], a major limitation is discussed of the technique requiring the introduction of SAPs, when very large core diameters are approached. Indeed

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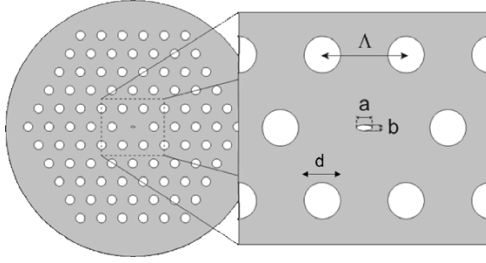


Fig. 1. Sketch of the transversal section of an EH-PCF. The fiber parameters are the hole pitch (Λ), the hole size (d), the air filling fraction (d/Λ), the longer elliptical hole axis (a), and the shorter elliptical hole axis (b).

the minimum distance between the SAPs and the inner fiber core dramatically increases when using larger PCF diameters, limiting the maximum achievable birefringence [19]. Moreover, this same problem would be even more severe if a SAP technique were applied to a double-clad PCF, given the presence of an extra air outer cladding. Thus, novel techniques and designs to realize PM-LMA-PCFs deserve to be investigated as an alternative or in addition to the SAP technique.

A novel highly birefringent fiber design was theoretically discussed in [20]. The proposed fiber was named “elliptical hollow optical fiber” (EH-OF) and was composed of a central elliptical air hole (with refractive index $n_1 = 1$), a circumferential elliptical ring germanium-doped core (with refractive index $n_2 > 1.45$), and a circular silica cladding. The birefringence of this fiber is related to the boundary conditions that apply to the electrical field at the interface between the central elliptical air hole and the concentric ring germanium-doped core. It is shown that, when the electrical field is normal to this interface, its magnitude in the germanium-doped core area (in the proximity of the air hole) is lowered by a factor depending on $\Delta n = n_2 - n_1$. Given the ellipticity of the air hole, orthogonal polarization fields, traveling in the doped-core area, experience such an effect over different interface extensions. This determines a difference in the two polarizations, i.e., birefringence [20].

On the basis of this same principle, we intend to demonstrate here that form-induced birefringence can efficiently be achieved also in the realization of LMA-PCFs. We design an elliptical hollow photonic crystal fiber (EH-PCF), including a central elliptical air hole, a concentric silica ring core and an air hole cladding (Fig. 1). We prove the feasibility of this fiber by fabricating three different EH-PCFs and characterizing their geometry and birefringence properties. Our numerical analyzes agree with the experimental results obtained and allow us to further investigate the dependence of the birefringence of an EH-PCF on some geometric features, as the elliptical hole dimension and ellipticity, or the cladding hole pitch Λ . We show that, by a simple modification of these geometric parameters, high values of birefringence can be obtained in EH-PCFs with large mode area.

II. EXPERIMENT

First, we have performed some numerical calculations by means of femlab, a commercially available software package based on finite element analyses, already used in the design of PCFs [21]. Our simulations showed, as expected, that, by

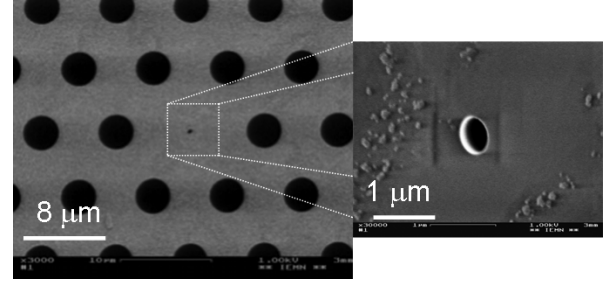


Fig. 2. SEM of our fabricated EH-PCF. A magnification of the central elliptical hole is shown in the inset.

choosing an air filling fraction $d/\Lambda < 0.43$, is possible to achieve single mode guidance in the ring core surrounding the elliptical air hole as well as form-induced birefringence.

In order to verify our predictions and prove the actual feasibility of an EH-PCF, we then fabricated three different EH-PCFs with large mode area. The fabrication of an optical fiber including an elliptical hole with dimensions less than $1 \mu\text{m}$ is a nontrivial problem and, to the best of our knowledge, has never been reported. One of the key points that we intend to demonstrate here is that an elliptical hole can be maintained while drawing it to a size of half a micron. The production of our fibers passed through two different stages. In the first fabrication step, we produced a silica cane comprising an elliptical hole in its center. In a second fabrication stage, we drew a final PCF preform containing in its center the cane previously made. A scanning electronic micrograph (SEM) of one of the fabricated fiber is shown in Fig. 2. A magnification of the inner elliptical air hole is shown in the inset. In Table I, we also show the measurements of different geometrical parameters of the produced fibers. As we can see, the ellipticity factor e of our fibers is comprised between 1.2 and 1.5 while the ellipticity factor of the starting silica cane that we have drawn was of 1.9. We believe that EH-PCFs with larger values of e can be obtained by using initial silica canes with a larger ellipticity and by optimizing the drawing conditions (temperature, pressure and speeds). As we can see in Table I, the calculated effective mode areas of our fibers A, B and C are, respectively, 52 , 100 , and $216 \mu\text{m}^2$. These values are larger than those obtained for the corresponding conventional PCFs, with the same hole and pitch parameters but without an elliptical hole in their center. In this case, the corresponding values of the effective areas would be 45 , 82 , and $183 \mu\text{m}^2$, respectively.

The guidance properties of the produced fibers were investigated through the use of white light, visible and infrared laser sources and of a CCD camera. A typical near-field mode image is shown in Fig. 3. Although our fiber geometric parameters do not respect exactly the condition for single mode operation [8], our observations confirmed the guidance of a single ring mode from the visible to the infrared regime. As observed in many other works, e.g., in [16], a very weak multimode guidance results in practice in a single mode behavior, due to the high losses experienced from the higher fiber modes.

We performed some group birefringence measurements on the produced fibers, using a standard crossed polarizer technique [22] with a supercontinuum laser source based on a highly

TABLE I
FIBERS GEOMETRY PARAMETERS AND CALCULATED EFFECTIVE MODE AREAS

FIBER NAME	FIBER DIAMETER $D(\mu\text{m})$	HOLE DIAMETER $d(\mu\text{m})$	HOLE PITCH $\Lambda(\mu\text{m})$	AIR FILLING FRACTION d/Λ	LONGER ELLIPTICAL HOLE AXIS $a(\mu\text{m})$	SHORTER ELLIPTICAL HOLE AXIS $b(\mu\text{m})$	ELLIPTICITY FACTOR $e=a/b$	HOLE ELLIPTICITY $1-b/a$	CALCULATED EFFECTIVE MODE AREA $A_{\text{eff}}(\mu\text{m}^2)$
FIBER A	91	2.7	5.9	0.45	0.405	0.270	1.5	0.33	52
FIBER B	123	3.6	8	0.45	0.590	0.425	1.39	0.28	100
FIBER C	194	6.5	12.8	0.5	0.525	0.425	1.23	0.19	216

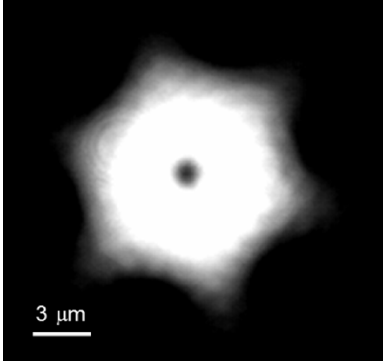


Fig. 3. Near-field mode image of “Fiber B” at a wavelength of 633 nm.

nonlinear PCF. Our experimental data are shown in Fig. 4, together with our numerical calculations. These data prove the existence of a form-induced birefringence in the produced fibers, which, at a wavelength of 1550 nm, extends from a value of about 1.2×10^{-5} in “Fiber C” to about 6×10^{-5} in “Fiber A.” As we can see, for all our measurements, there is a reasonable agreement between our numerical and experimental data. We believe their difference to be mainly related to structure imperfections and small errors in the measurements of fiber geometry parameters. In particular, the measurements of the dimensions of the elliptical axis of the central hole (a and b) have been taken by using our SEM images and are affected by a maximum error of about 15–20 nm. Although this number is small, it represents 3%–4% of the value of a and b and it leads to a maximum error of the ratio a/b of about 6%–9%. We have calculated that a small change in the choice of the hole ellipticity can determine a substantial modification of the fiber birefringence. Indeed, if we consider our fabricated “Fiber B” (Table I), a change of the ratio a/b of about 9% can induce a change in the fiber group birefringence of about 26% (at a wavelength of 1550 nm). In order to further outline this point, we have also shown in Fig. 4(b) the calculated group birefringence for a fiber with exactly the same parameters of “Fiber B,” but with the length of the longer and shorter hole elliptical axis modified of +10 and –10 nm, respectively. As we can see, this small modification already allows our calculated group birefringence to pass from a lower to a higher value of our measured birefringence. For both “Fiber B” and “Fiber C” our numerical data can fit our experimental data within an error limited to about 7%. Moreover the group birefringence of these two fibers increases with the wavelength [Fig. 4(b)–(c)], as in conventional PCFs [11]. In contrast we can

observe an uncommon behavior for “Fiber A” [Fig. 4(a)]. The group birefringence of this fiber presents a peak value at a wavelength of about 1500 nm and decreases at longer wavelengths. This unusual property is confirmed by our numerical data, although, in this case, they can fit our experimental data within a maximum error of about 16% at longer wavelengths. Again, we attribute this discrepancy to structure imperfections and geometry measurement errors, which can be particularly critical in the smaller core regime.

III. NUMERICAL ANALYSES

In order to explain the unusual behavior of “Fiber A,” to investigate the dependence of the group birefringence of an EH-PCF on fiber geometric characteristics, and to establish the eventual advantages of using an EH-PCF, we have performed some numerical analyzes. We have adopted an EH-PCF structure with the same geometric parameters of our “Fiber B” ($\Lambda_B = 8 \mu\text{m}$, $(d/\Lambda)_B = 0.45$, $a_B = 0.59 \mu\text{m}$, $b_B = 0.43 \mu\text{m}$, see Table I) and studied the behavior of its group birefringence by modifying its geometric features.

A. Wavelength Dependence

We can introduce a fiber magnification factor M to scale the fiber structure we have adopted. Thus, when $M = 1$ we refer to the fiber geometry of “Fiber B” while, in the other cases, we refer to an EH-PCF with a hole pitch $\Lambda' = M \cdot \Lambda_B$, a hole size $d' = M \cdot d_B$, and elliptical hole axis $a' = M \cdot a_B$ and $b' = M \cdot b_B$. The magnification parameter is particularly useful because it can describe the characteristics of different fibers with exactly the same structure but different fiber size.

We have shown, in Fig. 5, the dependence of the group birefringence on the wavelength for different values of the fiber magnification factor M . At short wavelengths, we can observe an increase of the birefringence for smaller fiber sizes. However, when adopting smaller M s, the peak value of the birefringence (corresponding to an “optimal wavelength” λ^*) shifts to shorter wavelengths. In order to understand this behavior, in Fig. 6 we show the difference between the principal polarization guided mode, for a fiber with $M = 0.75$, at different wavelengths ($\lambda_1 = 1.1 \mu\text{m}$ and $\lambda_2 = 1.8 \mu\text{m}$). As we can see, in Fig. 6(a) ($\lambda_1 = 1.1 \mu\text{m}$), the mode is completely guided in the surrounding silica core ring, leaving very low optical power inside the elliptical hole. In contrast, in the case of Fig. 6(b) ($\lambda_2 = 1.8 \mu\text{m}$), a longer light wavelength can penetrate better inside the hole (due to the diffraction effect), resulting in a more

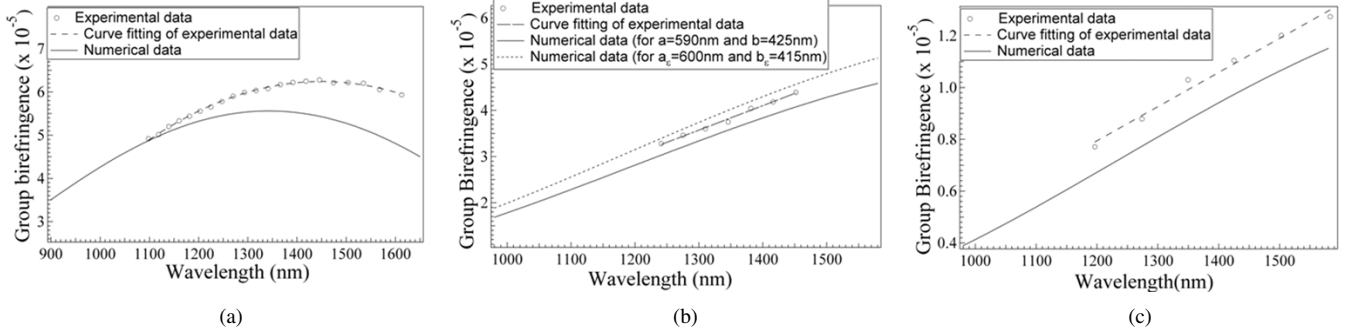


Fig. 4. (a) Group birefringence measurements of “Fiber A,” (b) “Fiber B,” and (c) “Fiber C.” A solid line indicates our numerical calculations. Our experimental data are represented by an empty circle ($\circ$) and their fitting curve is also shown. In (b), a dashed line above our experimental data indicates the calculated group birefringence of a fiber with the same pitch Λ and hole size d of “Fiber B” but elliptical hole axis modified of $+10$ nm (for the longer axis) and -10 nm (for the shorter axis).

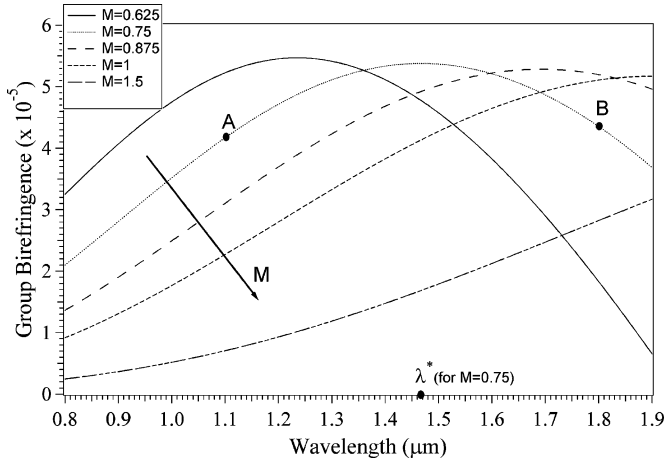


Fig. 5. Dependence of the group birefringence on the wavelength, for different values of the fiber magnification M . The marked points “A” and “B” indicate the value of the birefringence for $M = 0.75$ at a wavelength of 1.1 and $1.8 \mu\text{m}$, respectively. The point λ^* represents, for $M = 0.75$, the wavelength corresponding to a maximum of the fiber group birefringence.

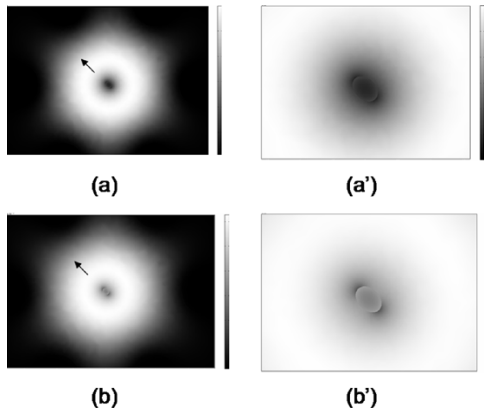


Fig. 6. Calculated image modes (on the main polarization axis) for a fiber magnification M of 0.75 at wavelengths of (a) 1.1 and (b) $1.8 \mu\text{m}$. The central elliptical air-hole is magnified in (a') and (b'). An arrow indicates the polarization direction and a graduation scale is shown on the side of the images, indicating increasing optical power levels from black to white.

significant portion of optical power located inside the central elliptical air-hole.

We can now show how an increasing percentage of the electrical field inside the hole can cause a decrease of the fiber birefringence. As explained extensively in [20], fiber birefringence

in an EOF is originated by the boundary conditions that apply on the electrical field at the air-silica interface

$$E_{\text{core},t} = E_{\text{air},t} \quad (1)$$

$$E_{\text{core},n} = \left(\frac{n_{\text{air}}}{n_{\text{core}}} \right)^2 \cdot E_{\text{air},n} \quad (2)$$

where E_{core} and E_{air} are, respectively, the electrical fields in the silica ring core and in the elliptical air-hole, n_{air} and n_{core} are the corresponding refractive indices, and the subscripts t and n are used to indicate, respectively, the electrical field components tangent and normal to the air-silica interface.

Thus, since E_{air} is almost zero and $(n_{\text{air}}/n_{\text{core}})^2$ is about 0.47 , relation (2) determines a depression of the electric field intensity in the silica ring core, in the proximity of the air-silica interface, because $E_{\text{core},n}$ is almost zero. The EH-PCF birefringence is due to the fact that orthogonal fiber polarizations experience this effect over different air-silica interface extensions [20]. However, when the electrical field intensity inside the hole E_{air} increases [as in the case of Fig. 6(b)], $E_{\text{core},n}$ increases (2) and the depression of the electrical field in the silica ring core, in the proximity of the air-silica interface, becomes less pronounced.

Thus, for wavelengths longer than an optimal wavelength λ^* , the percentage of light inside the central hole becomes significant, the boundary condition (2) determines a less important depression of the electrical field inside the ring core for both polarization modes, causing their disparity to be less significant. As a result, the fiber has a lower birefringence.

On the other hand, for wavelengths shorter than λ^* , the percentage of power inside the air hole is not significant and this effect should not be taken in account. By progressively decreasing the light wavelength, the optical mode power situated at the silica-air interface becomes less and less significant, leading to a lower birefringence.

B. Group Birefringence and Effective Area Tradeoffs

We have also studied the dependence of the group birefringence B_g on the fiber magnification M , at a constant wavelength of $1.55 \mu\text{m}$. In Fig. 7 we can see that, when M is less than about 0.8 , the birefringence decreases for smaller fiber sizes. This is a direct consequence of the same phenomenon explained above regarding the behavior of an EH-PCF at short wavelengths. Indeed

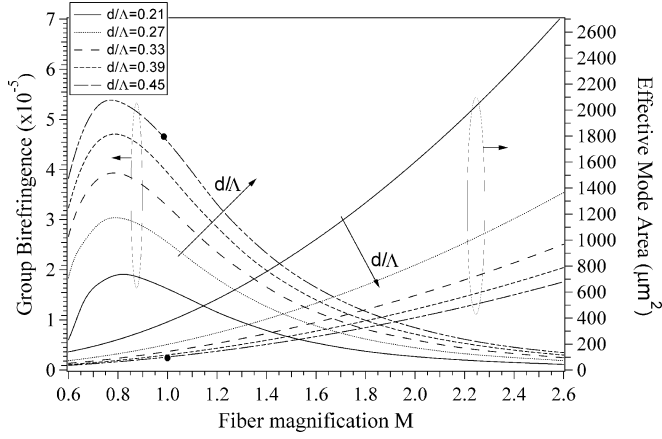


Fig. 7. Group birefringence and effective area tradeoffs. This figure shows the dependence of the group birefringence (on the left axis) and of the effective area (on the right axis) on the fiber magnification M , for different values of the air filling fraction and at a fixed wavelength of $1.55 \mu\text{m}$. Our fabricated fiber is indicated by a black spot (●, $M = 1$). As we can see a good compromise between high birefringence and large mode area is obtained.

the reduction of the fiber size determines an increasing optical power to be localized inside the central air hole. In contrast, if M is greater than about 0.8, the birefringence decreases for larger fiber dimensions. Since, in this same regime, the effective area increases, a compromise between high birefringence and large mode area should be obtained. Fig. 7 also shows the variations of the birefringence and effective area with respect to the air filling fraction d/Λ of the EH-PCF cladding. A compromise is also necessary in this case, because, while a large d/Λ leads to higher birefringence, it also causes a lowering of the effective mode area. Our fabricated “Fiber B” (Table I), also marked on the same plot, represents a good compromise among all these factors.

C. Elliptical Hole Size

It is important to establish the relation between the birefringence and the central elliptical hole size. First, this is useful to maximize the value of the birefringence and secondly is of paramount importance, from a fabrication point of view, to understand whether a small variation of the hole size would determine an abrupt decrease of the group birefringence.

For referring to the hole size, we have introduced a hole magnification factor m , such that $m = 1$ when the elliptical hole dimensions are the same than for the fabricated “Fiber B” (Table I). Thus, in general, the elliptical hole axis are $a' = m \cdot a_B$ and $b' = m \cdot b_B$ while the other fiber geometry parameters are fixed to the values $\Lambda' = \Lambda_B$ and $d' = d_B$. In Fig. 8 we have shown, at a fixed wavelength of $1.55 \mu\text{m}$, the relation between the group birefringence and m . We can observe a maximum value of the birefringence corresponding to an optimal hole magnification m^* . For values of m lower than m^* , light results more poorly confined in the concentric ring core and an increasing optical power is concentrated inside the hole, determining, as explained above, a lowering of the birefringence. However, for m greater than m^* , light confinement in the silica ring core becomes stronger as the central hole enlarges. In this case, as also discussed in [20], the percentage of optical power at the silica–air interface becomes less and less significant, leading to a lower birefringence.

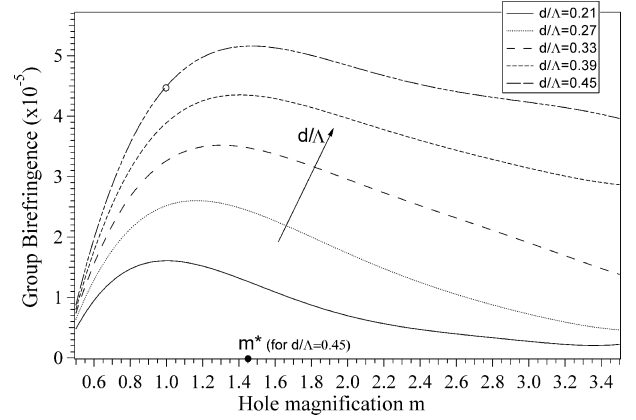


Fig. 8. Dependence of the group birefringence on the elliptical hole magnification m , for different values of the air filling fraction d/Λ , at a fixed wavelength of $1.55 \mu\text{m}$. The empty circle (○) represents the position in the graph of our fabricated “Fiber B.” The black spot (●) indicates the value m^* , for $d/\Lambda = 0.45$, at which the corresponding fiber birefringence reaches its maximum.

We also studied how the relation between the group birefringence and m changes with d/Λ . In Fig. 8 we can see that m^* is shifted toward higher values for greater d/Λ values. Indeed the optimal confinement of an optical mode inside the ring core requires a larger central hole size when a higher d/Λ is adopted.

From a fabrication point of view, this plot is very useful because it shows that, by choosing an appropriate elliptical hole size, a great level of tolerance in the fabrication is allowed for obtaining a high birefringence. For example, for $m = 1$ and $d/\Lambda = 0.45$, a value of the birefringence greater than 4.5×10^{-5} can be obtained for a range of m values between 1 and 2.45. In the same plot, we have also marked the position of our fabricated “Fiber B” ($m = 1$, $d/\Lambda = 0.45$).

D. Ellipticity

The birefringence of our EH-PCFs is due to the ellipticity of the central air hole. Therefore, it is of great importance to investigate the group birefringence dependence on the ellipticity value, in order to estimate which level of birefringence can be achieved in an EH-PCF.

We define an ellipticity factor “ e ,” such that the elliptical hole axis are $a' = e \cdot b'$ and $b' = b_B$. Thus, while the shorter elliptical axis is kept constant to the value of our fabricated “Fiber B” (Table I), the other axis is progressively lengthened, increasing the hole ellipticity (and dimension). Fig. 9 shows the relation between the group birefringence and the ellipticity factor e , for different values of the d/Λ (at a fixed wavelength of $1.55 \mu\text{m}$). As we can see, the increase of the birefringence with the ellipticity is more effective for fibers with a larger d/Λ ratio. Moreover, while the first three plots ($d/\Lambda = 0.45$, $d/\Lambda = 0.39$, $d/\Lambda = 0.33$) are characterized by a monotone increase of the birefringence, we can observe the presence of a maximum birefringence in the last two plots ($d/\Lambda = 0.27$, $d/\Lambda = 0.21$), corresponding to a value e^* of 4.2 and 3.1, respectively. When e is greater than e^* , the decrease of the birefringence is due to the enlarging of the hole dimension, which causes a stronger confinement of the optical mode in the silica ring core. As explained above, this can determine a lower birefringence because the amount of optical power at the silica–air interface is reduced.

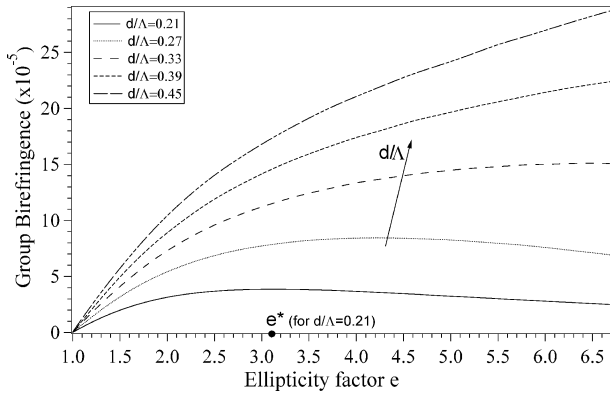


Fig. 9. Relation between the EH-PCF group birefringence and the ellipticity factor $e = a/b$, for different values of d/Λ , at a fixed wavelength of $1.55 \mu\text{m}$. The black spot indicates the value of the ellipticity factor e^* , for $d/\Lambda = 0.21$, corresponding to the maximum achievable birefringence.

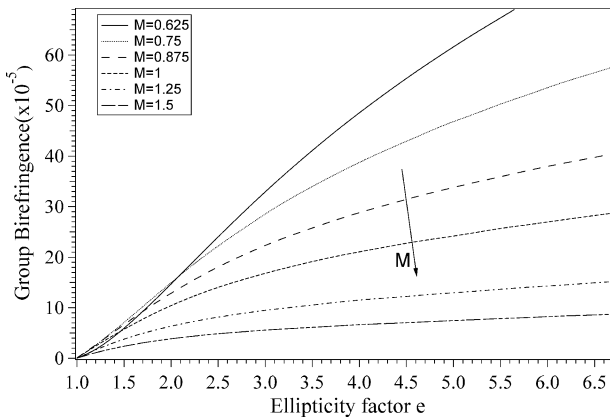


Fig. 10. Relation between the EH-PCF group birefringence and the ellipticity factor e , at a fixed wavelength of $1.55 \mu\text{m}$, for different values of the fiber magnification M .

Fig. 10 shows the same relation between the group birefringence and the ellipticity factor e , obtained for different values of the fiber magnification M . This is quite useful in order to determine which level of birefringence is possible to obtain in an EH-PCF. As we can see, for values of e greater than 3 and $M = 0.75$ (corresponding to a hole pitch $\Lambda = 6 \mu\text{m}$), a group birefringence greater than 2.85×10^{-4} is obtainable. Naturally much larger values of birefringence can be obtained by adopting a greater ellipticity or a smaller fiber size.

On the other hand, when considering very large mode area fibers, we can still achieve sufficiently high values of birefringence. For example, if we adopt $M = 1.25$ ($\Lambda = 10 \mu\text{m}$) and $e = 3.2$, we can obtain a group birefringence of 1×10^{-4} and, at the same time, an effective mode area of $167 \mu\text{m}^2$, greater than in commercially available LMA-PM-PCFs based on the SAP technique.

IV. CONCLUSIONS AND DISCUSSION

Highly birefringent fibers are of great interest for a wide range of applications and several techniques for their production have been developed over the years [15]. Given their particular guidance characteristics, the development of endlessly single-mode PM-PCFs with very large mode area is particularly attractive for their implementation in fiber devices, as lasers or gyroscopes.

Up to date, the only available way to obtain a highly birefringent LMA-PCF was based on stress-induced birefringence [16]. In this work we have investigated the possibility to obtain a form-induced birefringence in a LMA-PCF by inserting an elliptical air hole in its center. We have proved the feasibility of such a fiber (EH-PCF) and analyzed its birefringence properties and geometric characteristics both experimentally and numerically.

EH-PCFs may present an advantage over standard PANDA-PCFs in terms of smaller fiber size. Indeed in an EH-PCF there is no need for the insertion of SAPs in the outer fiber silica cladding. This makes EH-PCFs more practical and usable, and thus they could be an interesting alternative for applications in which the use of standard LMA-PM-PCFs would not be practical. Moreover the possibility of inserting SAPs in the silica outer cladding of an EH-PCF would combine the effects of stress and form-induced birefringence. This would result in a fiber with a higher birefringence, allowing the realization of larger core PM-PCFs.

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